

CHEMISTRY BOARD QUESTION PAPER : JULY 2023**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q.1 contains ten MCQs of one mark each. Q.2 contains eight very short answer questions of one mark each.
- (2) Section B: Q.3 to Q.14 contain twelve short answer questions of two marks each. (Attempt any eight.)
- (3) Section C: Q.15 to Q.26 contain twelve short answer questions of three marks each. (Attempt any eight.)
- (4) Section D: Q.27 to Q.31 contain five long answer questions of four marks each. (Attempt any three.)
- (5) Use of log table is allowed. Calculator not allowed.
- (6) Figures to the right indicate full marks.
- (7) For MCQs, correct answer must be written with its alphabet. Only first attempt will be evaluated.

Given: Avogadro Number $N_A = 6.022 \times 10^{23}$

SECTION – A [10]**Q.1. Select and write the correct answer for the following MCQs: [10]**

(i) Anisole on heating with conc. HI gives —

- (a) Iodobenzene (b) Phenol + Methanol (c) Iodobenzene + Methanol (d) Phenol + Iodomethane

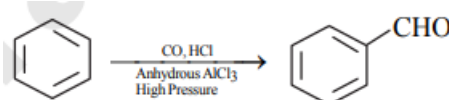
(ii) Which solution shows positive deviation?

- (a) Phenol–Aniline (b) Chloroform–Acetone (c) Ethanol–Acetone (d) Chloroform–Ethanol

(iii) Coordination number of Co in $[\text{CoCl}_2(\text{en})_2]^+$ —

- (a) 6 (b) 4 (c) 2 (d) 0

(iv) Name of reaction shown —



- (a) Etard (b) Friedel–Crafts acylation (c) Stephen (d) Gattermann–Koch

(v) Thermoplastic polymer —

- (a) Bakelite (b) Polystyrene (c) Nylon–6,6 (d) Urea–formaldehyde resin

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(vi) Nichrome is alloy of —

- (a) Cu–Sn (b) Cu–Ni (c) Ni–Cr (d) Fe–Cr

(vii) Identify A in reaction:



- (a) Bromobenzene (b) 1,4-dichlorobenzene (c) Naphthalene (d) Chlorobenzene

(viii) Amine that does NOT react with Hinsberg reagent —

- (a) Ethanamine (b) N-ethylethanamine (c) N,N-diethylethanamine (d) 2-methylpropan-2-amine

(ix) Dissociation constant of $\text{NH}_4\text{OH} = 1.8 \times 10^{-5}$. Degree of dissociation in 0.01 M solution

- (a) 0.04242 (b) 0.4242 (c) 0.004242 (d) 4.242

(x) Half-life of first-order reaction is 30 min at 300K. Rate constant K is —

- (a) 2.31 min^{-1} (b) 0.0231 min^{-1} (c) 0.231 min^{-1} (d) $2.310 \times 10^{-3} \text{ min}^{-1}$

Q.2. Answer the following: [8]

(i) Name radioactive element in group 16.

(ii) Structure of glycine.

(iii) Unit of cell constant.

(iv) Number of particles per unit cell in FCC.

(v) Name the γ -isomer of BHC.

(vi) IUPAC name of isobutyraldehyde.

(vii) Alloy used in Fischer–Tropsch process.

(viii) Three moles of ideal gas expand isothermally from 15 dm^3 to 20 dm^3 at 1.2 bar. Calculate work done (J).

SECTION – B

(Attempt any EIGHT) [16]

Q.3. Write four postulates of Werner's theory.

Q.4. Why does fluorine show anomalous behaviour?

Q.5. Mass of Cu deposited by 5A current in 6000 s ($\text{Cu} = 63.5 \text{ g/mol}$).

Q.6. Preparation of glucose from sucrose.

Q.7. Derive integrated rate law for zero-order reaction.

Q.8. Normal BP of ethyl acetate = 77.06°C . Solution: 50g solute in 150g solvent boils at 84.27°C ($K_b = 2.77^\circ\text{C kg mol}^{-1}$). Find molar mass.

Q.9. Preparation of phenol from cumene.

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Q.10. Why do d-block elements form coloured compounds?

Q.11. Write a note on Wolff–Kishner reduction.

Q.12. Preparation of Nylon–6,6.

Q.13. Derive Ostwald's dilution law for weak acid.

Q.14. What is Grignard reagent? How is it prepared?

SECTION – C

(Attempt any EIGHT) [24]

Q.15. Calculate ΔH° for reaction:

$\text{SiO}_2 + 3\text{C} \rightarrow \text{SiC} + 2\text{CO}$ using given data.

Q.16. Write a note on Hofmann bromamide degradation. Convert benzene diazonium chloride to benzene.

Q.17. Write three advantages & disadvantages of nanoparticles/nanotechnology.

Q.18. Give molecular formula & structure of: (i) H_2SO_4 (ii) H_2SO_5 (peroxy monosulphuric acid) (iii) $\text{H}_2\text{S}_2\text{O}_3$

Q.19. Explain optical activity of 2-chlorobutane.

Q.20. Write oxidation states of Mn. Explain why Mn^{2+} is most stable.

Q.21. Prepare using CH_3MgI : (i) ethanol (ii) propan-2-ol (iii) 2-methylpropan-2-ol

Q.22. Define ebullioscopic constant. Derive ΔT_f –molar mass relation.

Q.23. Define buffer solution. Write four applications.

Q.24. Element (27 g/mol) forms cubic unit cell with edge 405 pm, density 2.7 g/cm^3 . Identify nature of unit cell.

Q.25. Explain bonding in $[\text{NiCl}_4]^{2-}$ using VBT.

Q.26. Convert: (i) acetic acid $\rightarrow$ acetamide (ii) acetyl chloride $\rightarrow$ acetic anhydride (iii) sodium acetate $\rightarrow$ methane

SECTION – D

(Attempt any THREE) [12]

Q.27. Define isomorphism. Write Arrhenius equation and derive activation energy expression for temperatures T_1 and T_2 .

Q.28. What are interhalogen compounds? Write two characteristics. Draw Fischer projection of $\alpha\text{-D-(+)} \text{glucose}$. Write reaction for formation of Teflon.

Q.29. Describe construction & working of Standard Hydrogen Electrode. Write two difficulties in using SHE.

Q.30. Write two statements of first law of thermodynamics.

$\Delta H^\circ = -224 \text{ kJ}$ and $\Delta S^\circ = -153 \text{ J/K}$. Find temperature at which reaction becomes non-spontaneous.

Q.31. Define: (i) Gangue (ii) Ionization isomer (iii) Aromatic ketones

Write use and environmental effect of methylene chloride.

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